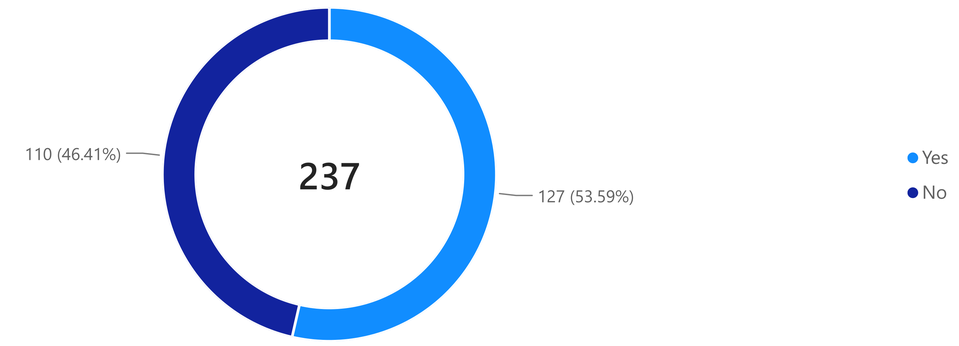
Total Employees
1470

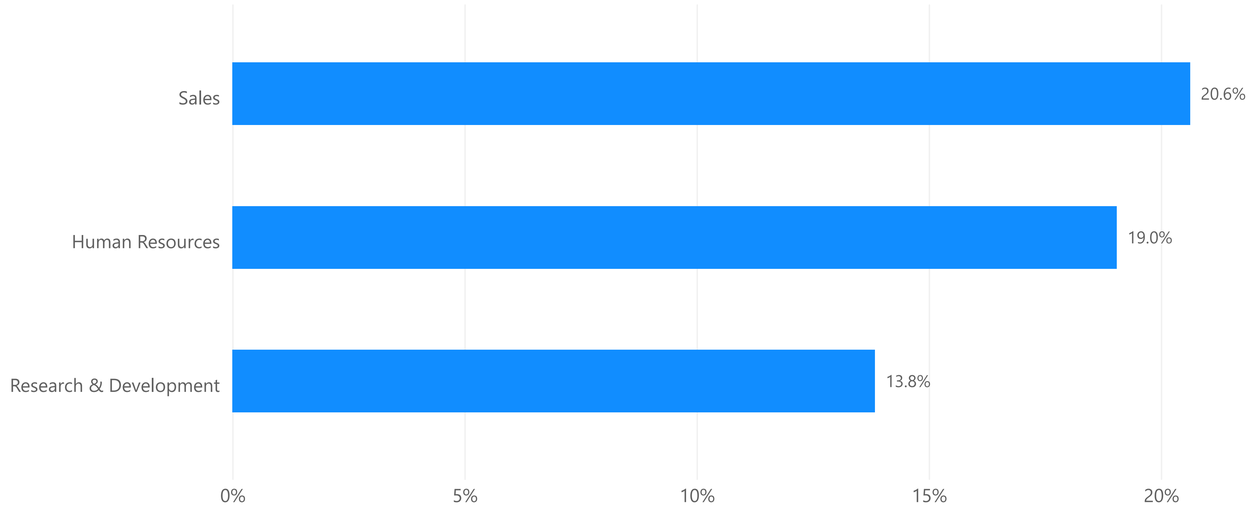
Total Attrition
237

Attrition Rate
16.1%

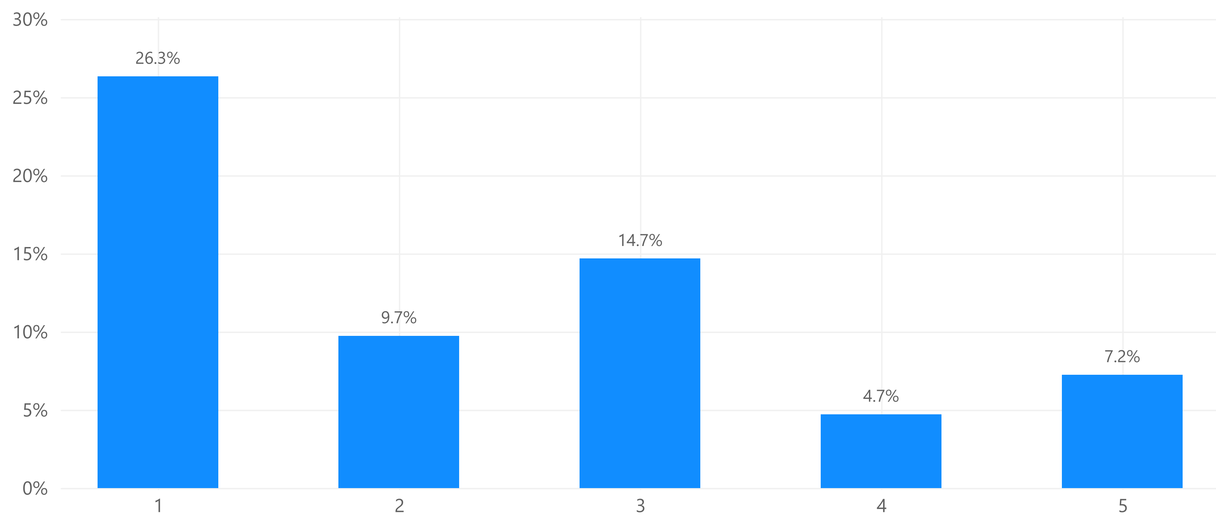
Total Attrition
by OverTime



Attrition Rate
by Department



Attrition Rate
by JobLevel



Department

Human Resources | Research & Develop... | Sales

JobLevel

1 | 2 | 3 | 4 | 5

OverTime

No | Yes

MaritalStatus

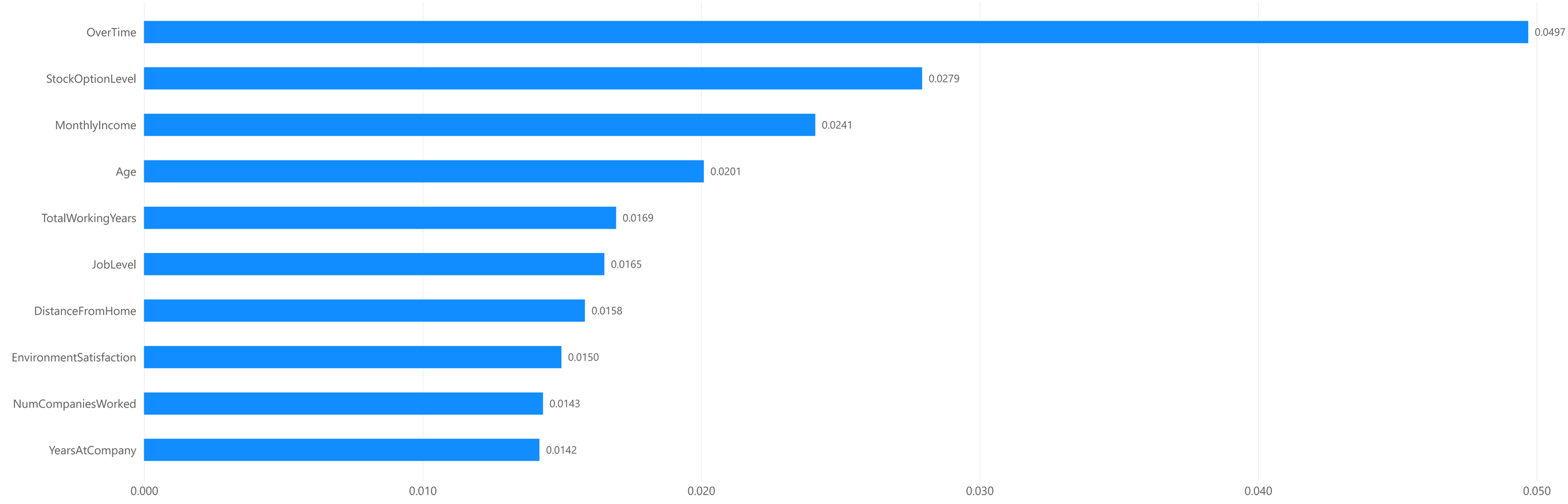
Divorced | Married | Single

Top Apriori Rules (Attrition Patterns)

Rank	Rule	Confidence_Pct	Support_Pct	Lift
1	Low income + New (2 yrs or less)	36.60	5.00	2.27
2	Works overtime	30.50	8.60	1.89
3	New (2 yrs or less)	29.80	6.90	1.85
4	Works overtime + Medium work-life balance	29.30	7.10	1.82
5	Young (under 30)	27.90	6.20	1.73
6	New + Medium work-life balance	27.60	5.40	1.71
7	Young + Medium work-life balance	26.70	5.00	1.66
8	Low income	26.10	8.70	1.62
9	Single	25.50	8.20	1.58
10	Single + Medium work-life balance	23.90	6.50	1.48

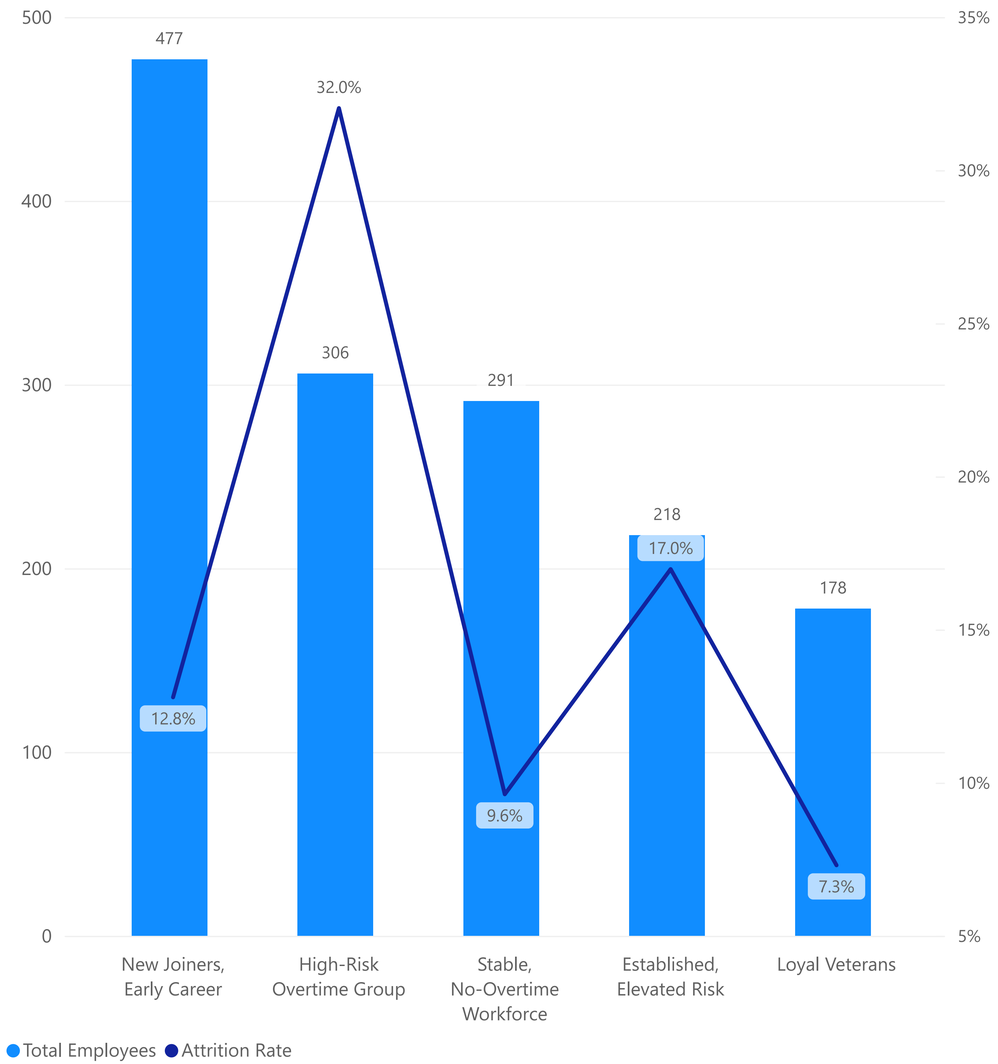
Top 10 Factors Driving Attrition (SHAP)

Sum of SHAP_Importance by Feature



Cluster Size and Attrition Rate

Total Employees and Attrition Rate by ClusterName



What do the five clusters mean for HR?

High-Risk Overtime Group (32.0% left): every employee here works overtime, and the group has the highest attrition. Cutting overtime workload is the clearest first action.

Established, Elevated Risk (17.0%): mid-career staff on mid-level pay, with about 29% working overtime. Watch workload and career progression.

New Joiners, Early Career (12.8%): the lowest pay and job level, with no overtime. Attrition is moderate, so pay and growth paths matter most here.

Stable, No-Overtime Workforce (9.6%): no overtime and steady tenure. This group is the benchmark for a healthy working pattern.

Loyal Veterans (7.3%): the oldest group, with about 16 years' tenure, the highest income and the highest job level. They are the most stable group, so HR should focus on retaining them.

Cluster Profiles

ClusterName	Total Employees	Attrition Rate	Average of Age	Average of MonthlyIncome	Average of YearsAtCompany	Average of JobLevel	Overtime %
Established, Elevated Risk	218	17.0%	36.38	5866.12	6.63	1.91	28.9%
High-Risk Overtime Group	306	32.0%	35.69	5324.34	5.53	1.78	100.0%
Loyal Veterans	178	7.3%	48.04	16043.51	15.88	4.21	26.4%
New Joiners, Early Career	477	12.8%	34.67	4840.31	5.46	1.70	0.0%
Stable, No-Overtime Workforce	291	9.6%	35.52	5108.84	5.96	1.76	0.0%

Department

Human Resources | Research & Develop... | Sales

JobLevel

1 | 2 | 3 | 4 | 5

OverTime

No | Yes

MaritalStatus

Divorced | Married | Single